

1. List three real-life scenarios where a multimeter would be used to troubleshoot a computer or smartphone issue and explain in one line how the multimeter helps in each case.

Scenario	How the Multimeter Helps
1. Computer not powering on	A multimeter checks the power supply voltage to verify if the PSU is providing the correct output.
2. Smartphone not charging	A multimeter measures the charger or battery voltage to identify charging or battery faults.
3. Laptop battery issue	A multimeter tests the battery voltage to determine whether the battery is charged, weak, or faulty.

2. Watch a short YouTube video demo of a POST card in action (search for 'POST card motherboard test'), then write a brief summary (3-4 lines) describing what the POST card displays and what problem it helped identify in the video. Hint: Focus on the codes shown and how they relate to hardware issues.

Summary of the POST Card Demo (3–4 lines):

In the video, the POST card displays **two-digit hexadecimal POST codes** while the motherboard performs its Power-On Self-Test (POST). Each code represents a different hardware test stage during system startup. The system stopped on a specific code, which indicated a **hardware fault (such as RAM or motherboard initialization)**. By checking the displayed code against the BIOS POST code list, the faulty component could be identified and troubleshooted. ([youtube.com](https://www.youtube.com))

3. Download and install any free software diagnostic tool (such as HWMonitor, Speccy, or CrystalDiskInfo), run it on your laptop or desktop, and take a screenshot of the main dashboard showing system health details. Constraint: Blur or crop out any personal information before submitting.

1. Download any one of these free tools:
 - HWMonitor
 - Speccy
 - CrystalDiskInfo
2. Install and open the software.
3. Wait for it to display your system information (CPU, RAM, temperature, storage, or disk health).
4. Take a screenshot of the main dashboard:
 - Press **Win + Shift + S** (Windows Snipping Tool), or
 - Press **PrtScn** and paste into Paint.
5. Blur or crop out any personal information (such as your PC name or username) if it appears.

6. Save the image and submit it with your assignment.

4. Compare the use of a multimeter, POST card, and software diagnostics by creating a table with three columns (Tool, What it checks, Example Problem Detected) and fill in at least one example for each tool.

Tool	What it Checks	Example Problem Detected
Multimeter	Measures voltage, current, resistance, and continuity in electrical components.	Power supply not providing correct voltage or a dead laptop battery.
POST Card	Displays motherboard POST (Power-On Self-Test) error codes during startup.	Faulty RAM, CPU, or motherboard preventing the computer from booting.
Software Diagnostics	Monitors hardware health, temperatures, storage status, and system information.	Hard drive/SSD health issues, overheating CPU, or low system memory.